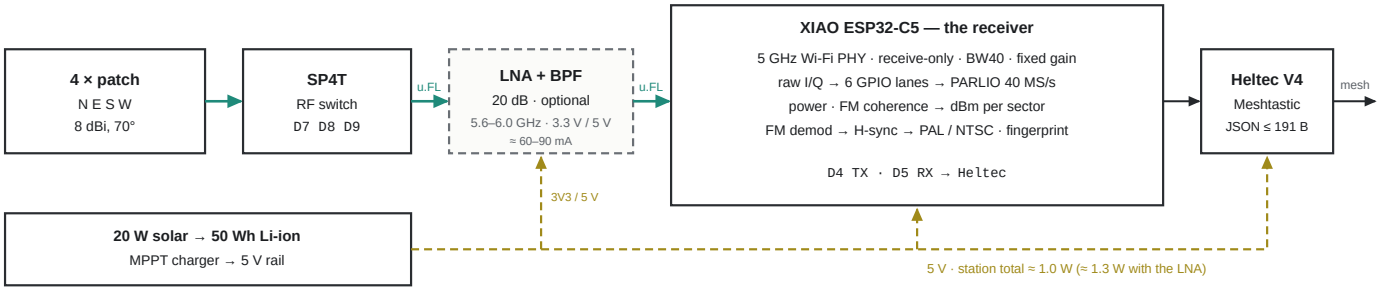
DRONE-SENTINEL · LEVEL 1 STATION · BENCH GUIDE

Level 1 Station v3 — C5 PHY prototype

The XIAO ESP32-C5's own Wi-Fi radio as the FPV receiver: sector direction finding on analog and digital video links, a software video check, no RX5808 and no sync separator. Bench build and verification guide.

Document	`docs/Level1-Station-v3-C5PHY-Bench-Guide.html`	**Firmware**	`level1-c5phy/`
Branch	`level1`	**Date**	2026-10-07
Status	**Both variants build and link (Arduino core 3.3.12 / ESP-IDF 5.5.5; RAM 45 %, flash 29 %), host tests pass; stage 0 passed on bench1 (2026-10-24), stages 1–3 started**	**Station**	

SIGNAL CHAIN AT A GLANCE



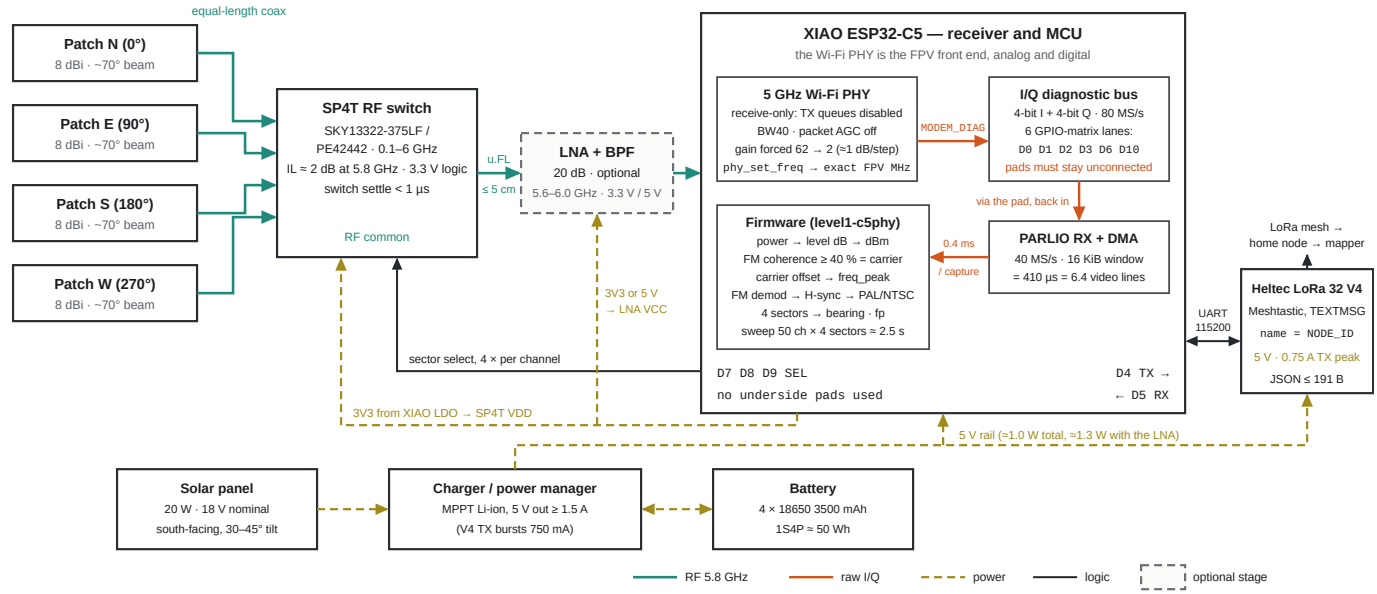
Simplified chain; the dashed LNA + band-pass stage is optional. The full block diagram with pin numbers, the I/Q path inside the C5 and the power tree is on sheet 2.

PARTS, AT A GLANCE

QTY	ITEM	KEY SPEC / PART	HAVE	≈ COST
1	XIAO ESP32-C5	receiver **and** MCU; only its U.FL antenna port is used	☐	$10
4	Directional patch antenna	5.8 GHz, 7–9 dBi, SMA; **gain entered in Station setup**	☐	$32
1	SP4T RF switch	SKY13322-375LF / PE42442	☐	$3
4	u.FL–SMA pigtail	**equal length**, ≤ 15 cm	☐	$8
4	SMA bulkhead feedthroughs	one per enclosure face	☐	$8
1	u.FL–u.FL pigtail	switch common → XIAO, ≤ 5 cm	☐	$2
1	LNA + band-pass filter (optional)	Optional 5.8 GHz, ~20 dB, 3.3/5 V; **single-band stations only**	☐	$12
4	Dual-band patch antenna	Dual-band 2.4–2.5 + 5.6–5.95 GHz, 6–9 dBi, SMA; instead of the 5.8 GHz ones	☐	$45
1	Wideband LNA, no filter	Dual-band 2.4–6 GHz, ~20 dB; instead of the filtered LNA	☐	$15
1	Heltec WiFi LoRa 32 V4	Meshtastic, TEXTMSG 115200, RX 47 / TX 48	☐	$20
1	Solar panel	20 W, 18 V nominal	☐	$25
1	Charger / power manager	MPPT, 5 V out, ≥ 1.5 A	☐	$15
4	18650 Li-ion	3500 mAh, 1S4P ≈ 50 Wh	☐	$24
1	IP65 enclosure + brackets	≈200×150×100 mm, 10–15° wedge	☐	$28
—	Passives	2 kΩ ×3, 100 nF ×2, 1000 µF	☐	$2
—	5.8 GHz VTX + camera (bench only)	NTSC **and** PAL cameras; a step attenuator; what each piece of bench kit is for: sheet 10	☐	—
0	Not needed any more	RX5808, LMH1980/LM1881 sync separator, video coupling parts, RX5808 pigtail	—	−$13

Before you start: bench1 passed stage 0; no transmitter has been tested yet. The receiver is the C5's Wi-Fi PHY driven through undocumented calls that the C5VRX project proved on ESP-IDF 6.0; this firmware re-implements them on the fleet's Arduino/ESP-IDF 5.5 toolchain. The procedure on sheets 6–9 is ordered so that each stage proves the next one is worth doing: receiver alive, channel coverage, selectivity, video (NTSC then PAL), sensitivity against an RX5808 station, the switch and the pattern, and last whatever digital video kit you have. Two build variants, one firmware: single-band (5.8 GHz patches, optional filtered LNA) and dual-band (dual-band patches, G1–G5 at 2.4 GHz added to the sweep); the sheets mark what the dual-band build adds. Two rules that are easy to break: the six I/Q lane pads (D0, D1, D2, D3, D6, D10) must stay unconnected, and every flash starts with a full erase. Full BOM on sheet 5.

1 · Full block diagram



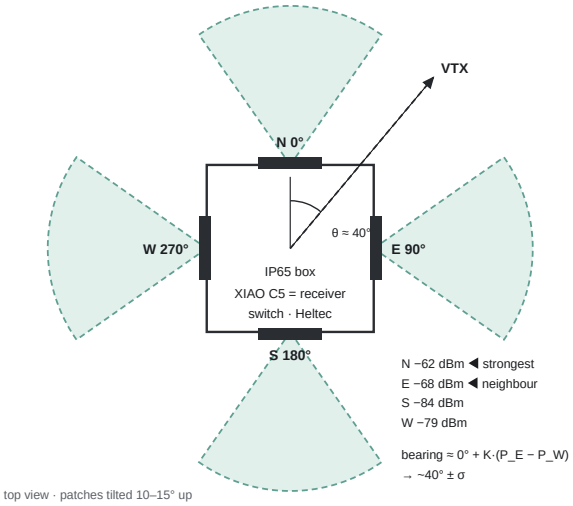
Four patches feed an SP4T switch whose common port goes into the XIAO's U.FL antenna connector, directly or through the optional 20 dB LNA + band-pass module. Inside the C5 the Wi-Fi PHY is held receive-only on the FPV frequency; its raw I/Q leaves on the modem's diagnostic bus, travels out through six top-side GPIO pads (bits 9–7 of I and Q) and back into the PARLIO peripheral, and the firmware measures power, FM coherence and horizontal sync on 16 KiB windows. The XIAO's 3V3 regulator feeds the switch (and the LNA, or the 5 V rail does); the 5 V rail feeds the XIAO and the Heltec. The LNA sits after the switch, so all four sectors share one gain: no per-sector gain mismatch, and the bearing constant K is unaffected.

WIRING LIST

FROM	TO	XIAO PIN	C5 GPIO	NOTES	✓
Switch V1	XIAO	`D8`	8	2 kΩ series	☐
Switch V2	XIAO	`D9`	9	2 kΩ series	☐
Switch V3	XIAO	`D7`	12	2 kΩ series; spare if the switch decodes 2 lines	☐
XIAO TX	Heltec V4 GPIO47 (Meshtastic RX)	`D4`	23	UART 115200 — the same pins as every station tier	☐
XIAO RX	Heltec V4 GPIO48 (Meshtastic TX)	`D5`	24	UART 115200	☐
Patches N/E/S/W	Switch RF1–4	—	—	**equal-length coax** ≤ 15 cm	☐
Switch common	XIAO U.FL	—	—	u.FL–u.FL ≤ 5 cm; the stock antenna comes off	☐
Switch common	LNA IN	—	—	Optional u.FL–u.FL or SMA pigtail ≤ 5 cm	☐
LNA OUT	XIAO U.FL	—	—	Optional ≤ 5 cm; replaces the direct switch → XIAO pigtail	☐
LNA VCC	3V3 or 5 V rail	—	—	Optional as the module requires; 100 nF + 10 µF at the module	☐
LNA GND	star point	—	—	Optional keep the module away from the Heltec's LoRa antenna	☐
I/Q lane pads Q7 Q8 Q9	**nothing**	`D0 D1 D2`	1 0 25	must float: no resistor, no probe, no PCB trace	☐
I/Q lane pads I7 I8 I9	**nothing**	`D3 D10 D6`	7 10 11	must float: same rule	☐
Underside pads	nothing	`GPIO2–5`	—	not used by this station: every pin is a top-side castellation	☐
5 V rail	XIAO 5V, Heltec 5V	—	—	1000 µF at the Heltec	☐
XIAO 3V3	Switch VDD	—	—	100 nF at the switch	☐
GND	all	—	—	single star point	☐

> **GPIO numbers are from the Arduino core's `XIAO_ESP32C5` variant** (`config.h`); wire by the D-labels. The I/Q lanes are GPIO 1, 0, 25 (Q7–Q9) and 7, 10, 11 (I7–I9): six of C5VRX's eight, the bit-6 lanes and the underside GPIO2–5 pads left out so the eleven top-side pins hold lanes, UART and switch; the firmware reads the missing bit as the midpoint (`IQ_LANE_BITS`). Each lane pad is driven and read back by the modem: any load on it corrupts the samples.

2 · Sector layout and receiver



Top view. Each patch covers about 70°, so an emitter always sits inside one sector and near the edge of a neighbour. Unchanged from v2.

METHOD

Amplitude comparison. Take the strongest sector `k` and its two neighbours; with powers in dB the bearing is

```
bearing = az[k] + K · (P[k+1] − P[k−1])
```

K (°/dB) is derived from the antenna gain entered at flash time (8 dBi → 72° beam → 2.97 °/dB, `bearing_k` in the boot line) and anchored by the stage 6 sweep. Report `bearing_deg` and `bearing_sigma_deg`; sigma grows on a sector boundary (strongest within 3 dB of a neighbour), on a weak peak, and when a neighbour's capture failed (sector axis, wide sigma).

PER-CHANNEL SCAN LOOP

```
rf_tune(channel); delay(8 ms)
for sector in N,E,S,W:
    select(sector); wait 200 µs; gain = 62
    for w in 1..3 (1 on a quiet channel):
        capture 16 KiB @ 40 MS/s     (0.4 ms)
        metrics                       (~2 ms)
        if clip > 3 %: gain -= 3; retake
    level = min(windows); q, cv2 = median
hit: best sector level >= 8 dB, q >= 40 %, cv2 <= 0.5
```

Sweep ≈ 50 channels × ~50 ms ≈ 2.5 s (55 dual-band ≈ 2.8 s) plus ≤ 0.5 s of confirmation passes. The gain steps keep strong carriers unclipped and weak ones at full gain; the minimum over windows drops Wi-Fi bursts.

HOW THE C5 BECOMES THE RECEIVER

STEP	WHAT HAPPENS	WHERE
Bring-up	Wi-Fi driver in station mode, power management off; 5 GHz only (both bands when dual-band), no power save, BW40; promiscuous with an empty filter; the five MAC transmit queues hardware-disabled; packet AGC off; PLL-tracking timer stopped. The radio cannot transmit.	`c5phy_rf.cpp` `rf_start`
Tune	Park on the nearest public 5 GHz centre the regulatory table accepts (`esp_wifi_set_channel`), then `phy_set_freq` to the exact FPV MHz; re-arm the diagnostic stream, re-assert AGC-off / BW40 / gain. Band A is exactly Wi-Fi ch 149–173.	`c5phy_rf.cpp` `rf_tune`
Capture	PARLIO reads the 6 lanes (bits 9–7 of Q and I) at 40 MS/s (every second modem sample) into one 16 KiB DMA window: 410 µs, one byte per sample, I high nibble, Q low nibble, the unwired bit-6 lanes read as the midpoint.	`iq_capture.cpp` `iq_capture_window`
Measure	Mean power; share of samples with power ≥ 8 whose phase step is within ±45° of the previous one (`q_phase`; an offset of ≤ 5 MHz); clipping; mean phase step → carrier offset; envelope variance / mean² (`cv2`); phase-step variance (video swing or bare CW).	`demod.c` `iq_metrics`
Level	`level_db = (62 − gain) × 1 dB + 10·log10(p_mean / 2.0)`; 2.0 is the no-signal power at gain 62 (on 3 lanes the decode floor, bench1). `rssi_dbm = −95 + level_db` until calibrated on sheet 7.	`main.cpp` `measure_sector`
Hit	Level ≥ 8 dB **and** coherence ≥ 40 % **and** `cv2` ≤ 0.5 on the strongest sector. Coherence is the selectivity: a carrier 10 MHz off steps 90° per sample and never counts, so a neighbour inside the 40 MHz filter is not a hit. `cv2` keeps OFDM (≈ 1) out; FM video reads 0.02–0.3.	`main.cpp` `scan_freq`
Video	On the strongest due peaks after the sweep: FM-demodulate 8 windows and look for horizontal sync repeating at the PAL or NTSC line period. Sheet 4.	`demod.c` `video_window` `main.cpp` `video_check`
Digital	Not a hit but continuous and noise-like (`cv2` 0.5–1.5, window spread ≤ 6 dB, each window 3 dB over the no-signal power): after the sweep, 8 windows on the strongest sector; lag autocorrelations class it `lte` / `dot11` / `wb`, `r1` gives the bandwidth bucket, the centroid the centre; duty ≥ 75 % → `"type":"wideband"` with a bearing. Sheet 9.	`demod.c` `iq_lag_*` `wideband.c` `main.cpp`

`demod.c`, `bearing.c`, `report.c` and `wideband.c` are plain C, unit-tested on a desktop with synthetic 4-bit I/Q (`test/host`, `make`).

3 · Video check and serial output

VIDEO CHECK

On the strongest due peaks after the sweep: hold the channel and the sector, demodulate 8 windows 5 ms apart (phase step between samples = instantaneous frequency), find sync-tip pulses 3.2–7.2 µs wide below a threshold set from the window's own percentiles, and measure the spacing of consecutive pulses.

LINE PERIOD	LINE RATE	VERDICT	MEANING
`64.0 µs (2560 samples)`	`15 625 Hz`	**PAL**	real analog video carrier
`63.56 µs (2542 samples)`	`15 734 Hz`	**NTSC**	real analog video carrier
sync-shaped pulses, no repeating period	—	**none**	not video-shaped: noise, Wi-Fi, an unmodulated carrier
swing under 4 units/sample	—	**none**	no FM deviation on this channel

A window counts with ≥ 2 consecutive periods and a sync score ≥ 70 (width closeness 0–40 + period closeness 0–60); the carrier is video when ≥ 3 of 8 windows count. `sync_q` is the share of windows that counted. The **field rate is not measured** (a 410 µs window never sees a vertical interval): `field_hz` is the nominal 50/60 of the detected standard. The fingerprint `fp` is `<video>/<sync_hz>/<freq_peak>`, e.g. `NTSC/15736/5734`, where `freq_peak` is the tuned frequency plus the measured carrier offset — no synthesizer sweep needed.

EXPECTED SERIAL OUTPUT

Boot line on USB at 115200. `"rf":true` is the PHY receive-only and the PARLIO reader running; `false` is followed by an `{"info":"error",…}` line naming the failed call. The bench console (`?` status, `h R3` / `h 5732` / `h D1` hold, `s 0–3` sector, `g 30` / `g a` gain, `v` video check, `w` wideband check, `k` / `k 0` stuck-window counts, `t 100` drive the switch lines while holding, `b 0|1` analog filter BW20/BW40, `x` resume) prints `{"info":"bench",…}` lines with `level_db`, `rssi_dbm`, `gain`, `p_mean`, `q_phase`, `cv2`, `clip`, `cfo_khz`, `mod`, `noise`, `step_std`, `nf_dbm`.

```
{"info":"c5phy v3 station ready","node_id":"A1B2","receiver":"c5phy","hw":"v3","channels":50,"sectors":
4,"heading":0,"fe_gain_db":0.0,"threshold_dbm":-87.0,"threshold_level_db":8.0,"q_min":40,"peak_pick":1,"v
ideo":1,"bw40":1,"gain_max":62,"window_us":409,"iq_lane_bits":3,"bands":"5.8","antenna_dbi":8.0,"beamwidt
h_deg":72,"bearing_k":2.97,"wideband":1,"pullin":1,"gain_step":3,"mesh_uart":"D4 TX / D5 RX 115200","phy_
set_freq":true,"rf":true}
```

Detection line on USB, one per hit per sweep — the v2 contract with `receiver`/`hw` changed and diagnostics added:

```
{"type":"analog_fm","receiver":"c5phy","hw":"v3","mac":"AF:00:52:03:16:64","freq_mhz":5732,"band":"R","c
h":3,"rssi":-68,"rssi_dbm":-68.1,"rssi_raw":536,"rssi_min":-70,"rssi_max":-66,"rssi_n":3,"level_db":26.
9,"gain":50,"q_phase":71,"cfo_khz":1840,"carrier":"fm","sectors":[-68.1,-74.4,-91.0,-85.2],"sector":0,"be
aring_deg":32,"bearing_sigma_deg":15,"station_heading":0,"freq_peak":5734,"video":"NTSC","sync_hz":1573
6,"field_hz":60,"sync_q":88,"sync_score":91,"video_windows":8,"fp":"NTSC/15736/5734","basic_id":"5.8G-R3-
5732MHz","node_id":"RX01","seq":42}
```

The mesh copy (Serial1 to the Heltec, ≤ 191 bytes, at most one per emitter per `MESH_REPORT_INTERVAL_MS`):

```
{"type":"analog_fm","mac":"AF:00:52:03:16:64","node_id":"RX01","freq_mhz":5732,"band":"R","ch":3,"rssi":-
68,"bearing_deg":32,"bearing_sigma_deg":15,"video":"NTSC","fp":"NTSC/15736/5734"}
```

Heartbeat every 60 s on USB, 120 s on the mesh:

```
{"heartbeat":true,"node_id":"RX01","receiver":"c5phy","hw":"v3","scanning":true,"channels":50,"sectors":
4,"heading":0,"threshold_dbm":-87.0,"threshold_level_db":8.0,"video_seen":7,"gain_max":62,"bw40":1,"fe_ga
in_db":0.0,"bands":"5.8","antenna_dbi":8.0,"beamwidth_deg":72,"bearing_k":2.97,"wb_seen":2,"pullin":3,"tu
ne_fail":0,"cap_err":0,"bus_stuck":0,"alias_drop":0,"sweeps":1830,"usb_drop":0,"mesh_drop":0,"nf_dbm":-9
8,"temp_c":41.2,"uptime_s":3600,"seq":42}
```

A digital video link is reported as `"type":"wideband"` under a `DF:` key, with `cls` (`lte` / `dot11` / `wb`), `conf`, `bw_mhz`, `fc_mhz`, `duty`, the lag features and the same bearing fields; the full USB and mesh lines are in `level1-c5phy/README.md`, the bench procedure on sheet 9.

> `rssi` is dBm like every other station tier; `rssi_raw` is the 10-bit RX5808-style value of the v2 contract. `bearing_deg` is relative to the box's face N — the mapper rotates it by the station heading set in its LEVEL 1 STATIONS panel, where each station's position is also set; two stations' crossing bearings give a position fix with an error radius. A station registers itself in the mapper from its first heartbeat. `tune_fail` counts channels the regulatory table refused, `cap_err` failed I/Q captures, `nf_dbm` the station's own noise-floor estimate from the quietest sector of the last sweep (`nf_dbm_24` for the 2.4 GHz plan on a dual-band build), `wb_seen` the digital-link reports, `pullin` the retunes onto off-channel analog carriers.

4 · Full bill of materials

Single-unit planning prices, quantities per station. Total ≈ $177 all-in, ≈ $189 with the optional LNA; ≈ $84 / $96 electronics only (v2: ≈ $190 / $95).

RF FRONT END

#	PART	QTY	NOTES	✓	≈ $
1	Seeed XIAO ESP32-C5	1	Receiver and MCU; only its U.FL port carries RF, the stock antenna is not used.	☐	10
2	Directional patch, 5.8 GHz, 7–9 dBi, SMA	4	**Enter its gain in Station setup** (K derives from it). Linear is polarisation-agnostic (−3 dB vs RHCP); RHCP +3 dB common case, −20 dB on LHCP. One batch, never mixed.	☐	32
	Dual-band patch, 6–9 dBi, SMA	4	2.4–2.5 + 5.6–5.95 GHz; replaces row 2 when dual-band; both gains entered.	☐	45
3	SP4T RF switch, 0.1–6 GHz, 3.3 V control	1	PE42442 (2/3-line decoded, drives directly) or SKY13322-375LF (4-line one-hot: needs a 3-input NOR for V4, see the README): ~2 dB loss at 5.8 GHz, sub-µs. Check control lines and truth table; eval board first (~$40).	☐	3
4	u.FL–SMA pigtails, equal length ≤ 15 cm	4	Matched length keeps sector losses matched — the bearing depends on it.	☐	8
5	u.FL–u.FL pigtail ≤ 5 cm	1	Switch common port to the XIAO's U.FL. Shortest run. Two with the LNA (switch → LNA IN, LNA OUT → XIAO).	☐	2
6	SMA bulkhead feedthroughs	4	One per enclosure face.	☐	8
7	Optional 5.8 GHz LNA + band-pass filter, ~20 dB; **single-band only**	1	~5.6–6.0 GHz passband, SMA or u.FL, 3.3 V or 5 V per module, ≈ 60–90 mA; between the switch common port and the XIAO's U.FL. Fit it when the stage 5 floor is worse than the v2 RX5808 station's or range is short; its filter also protects the front end from a nearby 5 GHz AP (the PHY's own filter is 40 MHz wide).	☐	12
	Wideband LNA, no filter	1	2.4–6 GHz, ~20 dB; replaces row 7 on a dual-band station; also hears every 2.4 GHz AP in view.	☐	15

MCU, MESH, POWER, MECHANICAL

#	PART	QTY	NOTES	✓	≈ $
8	Heltec WiFi LoRa 32 V4 + antenna	1	Meshtastic, TEXTMSG 115200, serial RX 47 / TX 48, node name = `NODE_ID`; D4/D5 as on every tier. Standard OLED model (the TFT model uses GPIO47/48). Fit the antenna before powering: it transmits at up to 28 dBm.	☐	20
9	Solar panel 20 W, 18 V	1	South-facing, 30–45° tilt. 10 W covers summer only.	☐	25
10	MPPT charger, 5 V out, ≥ 1.5 A	1	Must carry the Heltec V4's 750 mA transmit bursts on top of the ~0.2 A the rest of the station draws; a 1 A output is marginal.	☐	15
11	18650 Li-ion 3500 mAh, 1S4P holder	4	≈ 50 Wh. Protected cells or a BMS board.	☐	24
12	1000 µF low-ESR cap at the Heltec	1	LoRa TX bursts.	☐	1
13	IP65 enclosure ≈ 200×150×100 mm	1	Patches on the four vertical faces at 0/90/180/270°, tilted 10–15° up.	☐	15
14	Cable gland, desiccant	1		☐	3
15	Pole bracket + four patch wedges	1 set	Face N to true north and record the heading; a mis-oriented box rotates every bearing.	☐	10
16	2 kΩ ×3, 100 nF ×2	—	Switch control series resistors and supply decoupling (+ 100 nF and 10 µF at the LNA if fitted).	☐	1

REMOVED FROM THE V2 STATION

V2 PART	REPLACED BY
RX5808 receiver, SPI-modded	The C5's Wi-Fi PHY: public-centre bootstrap + `phy_set_freq` for tuning; raw I/Q power at a fixed gain for RSSI.
LMH1980 / LM1881 sync separator, 0.1 µF, 75 Ω	Software: FM discriminator on the same I/Q, H-sync search, PAL/NTSC from the line period.
RX5808 pigtail, 10 µF, SPI and sync wiring	Nothing. D0/D10 become I/Q lanes, D7/D8/D9 carry the switch.

> **Carrier PCB rules.** XIAO C5 footprint with D0, D1, D2, D3, D6, D10 **unrouted** (no test points; the underside GPIO2–5 pads are not used); SP4T with four u.FL launches and a u.FL to the XIAO (or a u.FL pair bridging the optional LNA); control lines on D7/D8/D9 via 2 kΩ; UART header on D4/D5; decoupling. Nothing else. Make it after stage 6.

5 · Bench procedure — is the receiver alive?

Work top to bottom on a bare XIAO with the stock antenna on the U.FL; the switch and patches come in stage 6. Do not move to the next stage until the current one gives the stated result. Console commands are typed into the serial monitor, Enter-terminated; the list is on sheet 4 and each stage names the ones it uses. What each piece of bench kit is for, stage by stage: sheet 10.

STAGE 0 — FLASH AND BOOT

- ☐ `pio run -e seeed_xiao_esp32c5 --target erase`, then `--target upload`. A **full erase precedes every flash**: stale PHY calibration data left C5VRX's receiver deaf after reflashes. Boot mode if it will not connect: hold BOOT, tap RESET, release BOOT.
- ☐ Monitor at 115200, then RESET: the line prints once. **Expected:** `{"info":"c5phy v3 station ready",…,"rf":true}`. If `"rf":false`, copy the `{"info":"error"}` line here: _____
- ☐ `?` twice, 60 s apart. **Expected:** `captures` climbing (about 100 a second while scanning), `cap_err` 0; `bus_stuck` a few % of the windows (bench1, known). Measured: captures/min _____ cap_err _____

STAGE 1 — RECEIVER ALIVE

- ☐ No VTX on. `h A1`. **Expected** (bench1, the decode floor): `level_db` 0.0, `p_mean` 2.00, `cv2` 0.00, `q_phase` 0, `gain` 62, `clip` 0, `noise` 0, `stuck` 0 (a dead I/Q bus reads `stuck` 1 on most lines; all-zero it also reads `p_mean` 2.0, so trust the flag). Measured: level_db _____ p_mean _____ q_phase _____
- ☐ Note the station's own no-signal floor: `nf_dbm` _____ dBm (bench1 −95). On 3 lanes `p_mean` 2.00 is the decode floor, not noise; another value means `RF_NOISE_POWER` needs this chain's figure. `cv2`, no signal: _____ (bench1 0.00).
- ☐ Key a VTX on A1 (5865 MHz) a few metres away. **Expected:** `level_db` up by ≥ 30 dB, `gain` stepping below 62, `q_phase` ≥ 50, `cv2` ≤ 0.3 (gate limit 0.5), `mod` 1, `noise` 0, `cfo_khz` within ±2000. Measured: level_db _____ gain _____ q_phase _____ cv2 _____ cfo _____
- ☐ Move the VTX away or unkey it. **Expected:** back to the no-signal values on the next line. `level_db` up without `q_phase`: an off-channel carrier (check `cfo_khz`) or scrambled lanes; with `cv2` ≈ 1 a digital signal or Wi-Fi (sheet 9).
- ☐ Dual-band `h G3`: the A1 values with no signal, `?` shows `band` 2; a phone hotspot raises the level. level_db _____ stuck _____

STAGE 2 — CHANNEL COVERAGE

- ☐ `x` to resume scanning; wait for a heartbeat (60 s). **Expected:** `tune_fail` 0, `channels` 50 (55 dual-band). Measured: tune_fail _____
- ☐ If not 0: `h` each band's channels and note the ones refused (`"err":"tune failed: esp_wifi_set_channel (regulatory table) (…)"`; L1–L3 on a `LOWBAND` 2 build fail with `no public centre within C5PHY_MAX_BOOTSTRAP_OFFSET_MHZ` instead, which no country table fixes): _____. Set `RF_COUNTRY_CC` to the deployment country, reflash (erase first), repeat. Value used: _____
- ☐ `h E4` (5645, lowest) and `h E8` (5945, highest). **Expected:** both hold; `?` shows the bootstrap `wifi_ch` 128 and 177. E8 is 60 MHz above the last public centre: if a VTX on E8 is not heard there but a VTX on E5 (5885) is heard on E5, E7 and E8 alike, the synthesizer does not follow `phy_set_freq` that far — build with `-DC5PHY_MAX_MHZ=5885` (46 channels) and note it here; `alias_drop` counts such mirrors, analog hits and digital candidates alike. Measured: _____ / _____
- ☐ `h D1`, `h L4`; dual-band `h G1`, `h G5`. **Expected:** all hold; `wifi_ch` 36 / 100 (G 1 / 13). No VTX on these Wi-Fi bands in most countries; an AP on Wi-Fi 36 (5180 MHz) is D1's test signal. Refused: _____ → `-DSCAN_5G1=0` / `-DLOWBAND=0`.

STAGE 3 — SELECTIVITY

- ☐ VTX on R3 (5732), scanning. **Expected:** a detection line for R3 within two sweeps; B1 (5733) and F1 (5740) may hit too and `PEAK_PICK` folds them into one. Channels reported: _____
- ☐ **Expected:** no hit on any channel more than 40 MHz away. If one appears, `h` it and note `q_phase` _____ (a hit needs ≥ 40) and `cfo_khz` _____.
- ☐ `h R3`, VTX on: `cfo_khz` _____ → `freq_peak` _____ MHz in the detection line. **Expected:** stable, the VTX's real offset.
- ☐ Hold the channel nearest a live 5 GHz access point (band A = Wi-Fi 149–173). **Expected:** raised `level_db`, `q_phase` under 40, `cv2` above 1.5: no hit of either kind. Under a long file copy a `wideband dot11` report with `duty` ≥ 90 is correct. cv2 _____ loaded: cls _____ duty _____

6 · Bench procedure — video and sensitivity

STAGE 4 — VIDEO CHECK

- [ ] VTX with an **NTSC** camera, hold its channel, `v`. **Expected:** ≥ 3 of 8 windows `video:1`, `std:NTSC`, `line_hz` 15734 ± 10, `present:1`. Measured: _____ / 8 windows, line_hz _____
- [ ] Same with a **PAL** camera. **Expected:** `std:PAL`, `line_hz` 15625 ± 10. Measured: _____ / 8, line_hz _____. **PAL is unproven on this receiver**: if it fails, record every window's `swing`, `pulses`, `periods` and `period`.
- [ ] VTX keyed with the camera unplugged. **Expected:** `present:0`; while scanning the detection line carries `"video":"none"` and `carrier` `cw` or `fm`.
- [ ] Back to scanning with the camera on. **Expected:** the detection line shows `video`, `sync_hz` and `fp`. Copy `fp` here: _____

STAGE 5 — SENSITIVITY AND DBM CALIBRATION

- [ ] VTX through a step attenuator (or a power meter on its output) into the U.FL; hold the channel and fill the level record below at each input level.
- [ ] **Expected:** `level_db` falls 1 dB per dB of attenuation over ~60 dB, `gain` rising back to 62 as the signal weakens, until the curve flattens at the floor; `q_phase` ≥ 40 and `cv2` ≤ 0.5 all the way down to the threshold. Measured slope: _____ dB/dB floor: _____ dBm
- [ ] Same VTX, same attenuator, on a v2 RX5808 station: floor _____ dBm. Difference v3 − v2: _____ dB. **This number decides whether v3 replaces v2 or supplements it.**
- [ ] Optional With the LNA fitted: repeat the attenuator sweep. Floor improvement v3+LNA − v3: _____ dB (expect 3–8 dB where the C5's own noise figure dominated); clipping onset at _____ dBm input; `RF_NOISE_POWER` measured: _____
- [ ] Solve the two-point line and enter `RSSI_CAL_DBM_AT_NOISE` _____, `RSSI_CAL_SLOPE` _____, `RSSI_CAL_OFFSET_DB` (per station, antenna + cable) _____; reflash. Heartbeat `threshold_dbm` now reads _____ (expect −85 to −90).
- [ ] Quiet band, 60 s of `h A1`: highest `level_db` seen _____ → set `DETECT_LEVEL_DB` at least 6 dB above it: _____
- [ ] **Filter skirt.** Attenuator set so `h R4` reads `level_db` 35–40: hold the channels 11–31 MHz off R4 in turn and fill the skirt record on sheet 10. It sets `WB_ANALOG_OWN_MHZ` (30 in the firmware, from a model of the skirt).

LEVEL CALIBRATION RECORD

Known level into the U.FL on a held channel. The slope should be 1 dB/dB; the two-point line sets `RSSI_CAL_DBM_AT_NOISE` and `RSSI_CAL_SLOPE` in `config.h`. Where the firmware has stepped the gain down, `level_db` already includes the `(62 − gain)` term. Dual-band: the same sweep on a G channel sets the `_24` set.

INPUT LEVEL (DBM)	−90	−80	−70	−60	−50	−40	−30
Measured `level_db`							
`gain` the firmware settled on							
`q_phase` (%)							
`cv2`							
Hit while scanning? (y/n)							

No transmitter: `level_db` _____ `p_mean` _____ `nf_dbm` _____. VTX used: _____ power / attenuator: _____ LNA fitted (y/n): _____

> **Reading the numbers.** `level_db` is dB above the noise reference (mean I²+Q² of 2.0 at gain 62); `rssi_dbm = RSSI_CAL_DBM_AT_NOISE + level_db × RSSI_CAL_SLOPE + RSSI_CAL_OFFSET_DB − RF_FRONTEND_GAIN_DB`. The default −95 dBm is kTB over ~20 MHz plus a ~6 dB noise figure, not a measurement. A slope far from 1 dB/dB, or a floor much worse than the RX5808's, is a real result — record it, do not tune the constants to hide it.
>
> Optional **With the LNA fitted**, run this calibration with the LNA in the chain and leave `RF_FRONTEND_GAIN_DB` at 0; `RF_FRONTEND_GAIN_DB = 20` only carries a calibration taken without the LNA over to a station that has one. Re-measure `RF_NOISE_POWER` (`p_mean` with no signal at gain 62): the LNA's output noise can exceed the C5's own floor. Expect the gain stepping (gain index 62 → lower) to engage ~20 dB earlier.

7 · Bench procedure — switch, pattern, field, power

STAGE 6 — RF SWITCH AND PATTERN

- [ ] Eval board powered from 3V3, one patch on RF1, common port → XIAO U.FL, control lines from D7/D8/D9 through 2 kΩ. Hold a channel with a VTX on, drive each combination (`t 000`, `t 100`, `t 010`, `t 110`; the pattern stays until `s` or `x`) and record which RF port goes through. **Write the truth table here** (the firmware's V1 is D8, V2 D9, V3 D7): `V1V2V3 = 000 → RF__  100 → RF__  010 → RF__  110 → RF__` → `SECTOR_SWITCH_TABLE`
- [ ] VTX on, hold its channel, `s 0` … `s 3`. **Expected:** `level_db` follows the sector the patch is on and settles within one line. Insertion loss against the patch wired direct to the U.FL: _____ dB (expect ≈ 2); measured with the LNA in the path if fitted: _____ dB (expect ≈ −18, a net gain)
- [ ] Mount all four patches on the box, VTX at 30 m on a known bearing, clear line of sight. Rotate the box in 15° steps through a full turn, logging all four `sectors` values at each step.
- [ ] Derive K (°/dB) from the neighbour difference: _____; the boot line's derived `bearing_k`: _____ (2.97 at 8 dBi). Record the measured K as `custom_bearing_k` and `BEARING_K_SCALE` = 2.5 × measured / derived: _____. Save the sweep to `docs/`.

STAGE 7 — FIELD ACCEPTANCE

- [ ] Two v3 stations 300–500 m apart, both positions and face-N headings set in the mapper's LEVEL 1 STATIONS panel, both named to match their Meshtastic nodes, D4/D5 wired to the Heltecs.
- [ ] Walk a VTX between them on ten marked points. **Pass:** the 90 % circle contains the true position on at least 9 of 10. Score: _____ / 10
- [ ] **Pass:** `video` reads NTSC or PAL on every report for a live camera, never `none`; the mesh copy arrives with `fp`.
- [ ] Leave both stations on solar for 72 h. **Pass:** no brownout, heartbeat gap never exceeds 5 min. Measured station current at 5 V: _____ mA (expect ≈ 200–240; ≈ 270–320 with the LNA, and peaks near 0.9 A while the Heltec transmits)

POWER BUDGET

LOAD	CURRENT	POWER	NOTES
XIAO ESP32-C5, Wi-Fi PHY receiving continuously, 240 MHz	~120 mA @ 3.3 V	~0.45 W	via the 5 V rail and on-board LDO. The radio never sleeps and never transmits; PARLIO/DMA add little. Datasheet estimate — the stage 7 current measurement replaces it
SP4T switch	< 5 mA	0.02 W	the entire DF add-on
Optional LNA + filter (optional)	60–90 mA @ 3.3/5 V	≈ 0.3 W	fit only if stage 5 shows it pays
Heltec V4, Meshtastic, mostly RX	75 mA RX, ~110 mA avg @ 5 V	≈ 0.55 W	750 mA (3.75 W) while transmitting at 27 dBm (Heltec V4 datasheet); the average assumes 5 % airtime
Station total		**≈ 0.9–1.1 W, ≈ 1.2–1.4 W with the LNA**	**≈ 22–34 Wh/day; peak ≈ 0.9 A at 5 V (≈ 1.0 A with the LNA) while the V4 transmits, so the 5 V output needs ≥ 1.5 A.** The v2 solar sizing (20 W panel, 50 Wh battery, ~36 h of darkness at 1.4 W) still carries it, with less margin; a 10 W panel is then summer-only. The RX5808's 0.6 W is gone and the radio's ~0.4 W takes its place

RECORD

Built by		**Date**	
Verified by		**Date**	
Station ID		**True-north heading of face N**	
Firmware commit		**Platform / libphy checked**	

Copy every stage's numbers into the station record; the v2 guide stays the reference for the truth table, pattern sweep and field details.

8 · Bench procedure — digital video links

Per kit you have: an OpenIPC / wfb-ng link, HDZero, DJI O4 with its goggles, Walksnail Avatar — a running link a few metres from the box on a channel the kit shows. The station must keep it out of the analog gate, confirm it as `"type":"wideband"` and class it; the mapper (level2-main or standalone, both carry the classification table) must name the system. No kit has been on the bench yet: the thresholds come from a synthetic model of 802.11 and LTE-like waveforms, and HDZero and Walksnail have no model at all — for them the numbers are the result, not a pass mark. Build / gains entered: _____ / _____ / _____ dBi

STAGE 8 — DIGITAL LINKS

- ☐ Kit on; `h` the table channel nearest its centre (*ch held* below), `s` the sector facing it. **Expected:** `level_db` well above 8 dB, `q_phase` under 40 (near 40 on a DJI link), `cv2` 0.5–1.5: the analog gate rejects it, the candidate rule takes it. Measured: level_db _____ q_phase _____ cv2 _____
- ☐ `w` three times (the 8-window pass plus a lag scan 40–3200 of the last window, about 2 s each). **Expected:** `duty` ≥ 75, a `cls` that fits the kit, `bw_mhz` and `fc_mhz` matching its setting, the same answer on every press (`scan_lag` may miss 2665–2669 on one DJI press in five). Fill the kit's block.
- ☐ `x` with the mapper open. **Expected:** one `wideband` line per sweep under a `DF:` key, no `analog_fm` line for the channel, `wb_seen` climbing in `?`; the mapper's IDENTIFICATION panel names the system with a confidence (`system`, `system_conf`). Record the label.
- ☐ Kit at 30 m on a known bearing, the box rotated 0°, 15°, 30°, 45° off the sector axis as in stage 6. **Pass:** |true − `bearing_deg`| ≤ `bearing_sigma_deg` at every step (≈ 15° at duty 100, 20° at duty 75: one window per sector, interleaved N E S W W S E N). The neighbour-difference slope against the angle is the digital K; the firmware uses one K per band for analog and digital alike, so a K unlike stage 6's is a finding to report, not a constant to enter.

RECORD — ONE BLOCK PER KIT

OPENIPC	CH HELD	CLS	CONF	BW_MHZ	FC_MHZ	DUTY	R128	R2667	SCAN_LAG	SYSTEM
expected	A7…A1	dot11	high	20 (40)	centre	≥ 90	≥ 0.10	~0.01	128	802.11 video link
measured										
bearing	0°: _____ 15°: _____ 30°: _____ 45°: _____ σ: _____ digital K: _____ °/dB kit set: _____									

HDZERO	CH HELD	CLS	CONF	BW_MHZ	FC_MHZ	DUTY	R128	R2667	SCAN_LAG	SYSTEM
expected	R1–R8	any	med/low	30 / 20	centre	≥ 75	record	record	record	HDZero
measured										
bearing	0°: _____ 15°: _____ 30°: _____ 45°: _____ σ: _____ digital K: _____ °/dB kit set: _____									

DJI O4	CH HELD	CLS	CONF	BW_MHZ	FC_MHZ	DUTY	R128	R2667	SCAN_LAG	SYSTEM
expected	R4 · B4 · B5	lte	high	10/20/40	centre	≥ 75	< 0.05	≥ 0.03	2665–2669	DJI O4
measured										
bearing	0°: _____ 15°: _____ 30°: _____ 45°: _____ σ: _____ digital K: _____ °/dB kit set: _____									

WALKSNAIL	CH HELD	CLS	CONF	BW_MHZ	FC_MHZ	DUTY	R128	R2667	SCAN_LAG	SYSTEM
expected	R1–R8	any	med/low	20 (40)	centre	≥ 75	record	record	record	Walksnail Avatar
measured										
bearing	0°: _____ 15°: _____ 30°: _____ 45°: _____ σ: _____ digital K: _____ °/dB kit set: _____									

ch held: the table channel nearest the kit's centre — Band A is Wi-Fi 149–173 (A7 = 149 … A1 = 173), D1–D3 cover Wi-Fi 36–48, HDZero's and Walksnail's channels sit within 4 MHz of R1–R8, DJI O4's 5768.5 / 5789.5 / 5794.5 / 5814.5 MHz fall to R4 / B4 / B4 / B5 and a CE unit to D1–D3. `fc_mhz`: the kit's centre. `conf` is `high` at duty ≥ 90 with the deciding feature clear of its threshold, `med` otherwise, `low` when `cls` is `wb`. A 60 MHz O4 mode reads as the 40 bucket. *bearing*: true − `bearing_deg` at each rotation, σ = `bearing_sigma_deg` reported, kit set = the channel / bandwidth selected on the kit.

> `Dual-band` **2.4 GHz.** Repeat the block for any kit with a 2.4 GHz mode (DJI OcuSync 2 / DJI FPV, an OpenIPC link on a 2.4 GHz adapter): `h G1`–`h G5`, the same `w` and rotation. Under test are the patches' 2.4 GHz side, `ANTENNA_GAIN_DBI_24` (6 dBi → 90° beam, K 4.7 °/dB) and the `_24` calibration from stage 5; the mapper labels it a 2.4 GHz digital video link. Expect every 2.4 GHz access point in view as `dot11` traffic (idle: duty under 90, "Wi-Fi traffic"). I/Q at BW40 on 2.4 GHz is still unproven (stage 1, `h G3`).

9 · Bench kit — what each piece is for

Against the stages of sheets 6–9: Tier 0 of the shopping list (VTX, cameras, attenuator, router) carries stages 0–7; Tier 1 (the digital kits, in purchase order) is stage 8, one record block per kit; Tier 2 is the dual-band build's repeat of stages 1, 5, 6 and 8. In stage 8 keep any analog VTX off, or more than 30 MHz from the kit's channel: an analog carrier within `WB_ANALOG_OWN_MHZ` owns the digital candidates near it, by design.

KIT	STAGE	WHAT YOU DO WITH IT	WHAT IT SETTLES
Analog 5.8 GHz VTX, one NTSC and one PAL camera	1, 3, 4, 5, 6, 7	A1 keyed and unkeyed; R3 with B1 / F1 folding; `v` per camera; the attenuator sweep; the 30 m pattern sweep; the ten-point walk	Hearing at all; `PEAK_PICK`; PAL; `RSSI_CAL_*`, `DETECT_LEVEL_DB`, the v2 / v3 verdict; K; the fix circle
Step attenuator (or a power meter)	5	Between the VTX and the U.FL: the level record, then the skirt record below	Slope, floor, the two-point line; `WB_ANALOG_OWN_MHZ`
5 GHz Wi-Fi router + laptop	2, 3	An AP on Wi-Fi 36 for D1; its channel held idle, then under iperf or a long file copy	Idle Wi-Fi never reported; a saturated link reads `dot11`, `duty` ≥ 90
OpenIPC / wfb-ng link, two RTL8812AU adapters	8	The OpenIPC block: `h` its channel, `w` three times, `x` with the mapper, the rotation	Whether the C5 decodes its frames at all; `WB_R128_MIN`; the digital K
HDZero VTX + camera	8	The HDZero block	Its waveform: `cls` and bucket are the result
DJI O4 Air Unit + Goggles N3	8	The DJI O4 block on R4 / B4 / B5 (a CE unit on D1–D3); the goggles later, for pilot presence	`WB_R2667_MIN`, the `lte` class, the O4 row's confidence
Walksnail Avatar VTX + VRX	8	The Walksnail block	Class and bucket on 5770 / 5839
Dual-band patches ×4	6, 8	The pattern sweep on the dual-band build; stage 8's 2.4 GHz block	`custom_bearing_k`; the `_24` gain and K
2.4 GHz analog VTX	1, 5	Keyed on G3 instead of a hotspot; the attenuator sweep on a G channel	The I/Q bus at 2.4 GHz; the `_24` calibration
Wideband LNA (dual-band) or 5.8 GHz LNA + filter (single-band)	5, 6	The optional LNA sweep; its gain in the switch test	Fit it or not; `RF_NOISE_POWER`, `RF_FRONTEND_GAIN_DB`
Borrowed DJI drone, OcuSync 3 / 4	8	The 2.4 GHz block	The 2.4 GHz digital path; the OcuSync rows
Patches, SP4T evaluation board, pigtails	6	Truth table with `t`, `s 0`–`s 3`, insertion loss, the pattern sweep	`SECTOR_SWITCH_TABLE`; K

STAGE 5 ADDENDUM — THE CHANNEL FILTER'S SKIRT

A strong FM carrier's deviation swings across the edge of the 40 MHz channel filter; the skirt turns that into amplitude modulation, and a table channel beyond the edge sees a non-coherent, noise-like, steady signal: the wideband candidate rule. The firmware lets an analog carrier own the candidates within `WB_ANALOG_OWN_MHZ` 30 of it, a value set on a model (a soft skirt passes the rule at 22–24 MHz, a sharp one at 18). This measures the real one.

- ☐ VTX on R4 (5769) through the attenuator, set so `h R4` reads `level_db` 35–40. Hold each channel of the record, note `level_db`, `q_phase`, `cv2`. **Expected:** `q_phase` under 40 everywhere, the level falling with the offset. A channel with `level_db` ≥ 8 **and** `cv2` 0.5–1.5 is one the candidate rule would take: the largest such offset plus 5 MHz is the radius needed. Over 30 → raise `WB_ANALOG_OWN_MHZ`; nothing beyond 20 → 25 is enough. Radius: _____ MHz
- ☐ `x`, VTX still on R4, two minutes. **Expected:** no `wideband` line, `wb_seen` in `?` steady: the carrier owns its images. A `DF:` line more than 30 MHz from R4: note it _____

HELD CHANNEL (OFFSET FROM R4)	F3 5780 (+11)	A5 5785 (+16)	B4 5790 (+21)	A7 5745 (−24)	F1 5740 (−29)	F4 5800 (+31)
`level_db`						
`q_phase` (%)						
`cv2`						
Would pass the candidate rule? (y/n)						

`h R4` level_db _____ attenuator _____ dB VTX / camera: _____ LNA fitted (y/n): _____. Model for comparison (soft skirt, 30 dB over threshold): +22 → level 20, cv2 0.7; +24 → 14, 1.4; +26 → 8, 2.0 (fails on cv2).